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Optimization of twisted-tape element static mixers for industrially relevant setups

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ABSTRACT

Static mixer studies in literature have often been reported for idealized systems with a concentric injected flow. These do not account for real-world mixing scenarios where the additive comes in through an injector. Depending on the injector geometry, the injected flow, and pipe Reynolds number, the injected flow may end up off-centered. This impacts the static mixer performance as the first few elements are unable to provide efficient mixing. This paper discusses a parametric study of twisted-tape mixers to numerically ascertain their performance for Newtonian flows. It focuses on industrially relevant mixers which operate with some inexactness, e.g., off-center flow of the additive. Effect on mixing of angle of twist and aspect ratio of the individual element along with Reynolds number are studied; and the Reynolds number dependency of mixer geometry for optimal performance is established. This is done without using some oft-used approximations/assumptions in previous literature like zero-thickness mixer walls, 2d analysis, periodic boundary conditions etc. Two non-dimensional quantities, i.e., Coefficient of Variation (CoV) and Z-factor, are used to assess and quantify the operation of the mixers. For idealized systems, increasing the number of elements by reducing the aspect ratio and/or the twist angle is seen to improve mixing for all Reynolds numbers tested (1–100). However, for a real-world mixer constrained by its total length/pressure drop, mixing is proven to be an optimization problem between increasing the number of flow-splits by introducing a greater number of elements and mitigating the anomalous flow (uneven split) caused due to it. For $Re \leq 10$, reduction of twist angle from 180° to 140° improves mixing for only aspect ratios greater than approximately 0.9. For Reynolds number of 100, the 140° twist angle outperforms the 180° twist angle for all aspect ratios studied. In summary, there is no combination of twist angle and aspect ratio that is a universally optimal solution and these parameters must be decided subjectively based on multiple other input parameters for a given system.

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1. Introduction

Mixing is an important consideration in petrochemical, plastic, textile, and other industries. Insufficient mixing can affect the quality of the product by promoting side reactions and/or leaving inhomogeneous pockets; which, in turn, result in economic losses and undesirable impact on the

environment (waste). Mixing is attained via either a batch or a continuous process. For the latter, static mixers or motionless mixers are an important class of devices. “Static mixer” is a generic term for a wide variety of equipment used to achieve a broad range of mixing applications. They have specially shaped inserts to redirect the flow to enhance shear and mixing. The principle of operation is splitting or cutting of the flow and its recombination in a reordered configuration. This creates layers which grow progressively thinner and more in number as the flow propagates downstream (Paul et al., 2003; Thakur et al., 2003). Desired process results

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Nomenclature

KSM	Kenics Static Mixer.
θ	Angle of twist of an element.
ρ	Density of a fluid.
σ_C	Standard deviation of additive concentration along a cross-section.
μ	Dynamic viscosity of a fluid.
AR	Aspect ratio of an element ($= L/D$).
C_{av}	Mass-weighted average additive concentration along a cross-section.
C_f	Friction factor.
CoV	Coefficient of variation.
D	Diameter (inner) of pipe.
D_i	Mass diffusivity for a fluid pair.
L	Pitch of an element ($= L_a$ if $\theta = 180^\circ$).
L_a	Actual length of an element.
L_{tot}	Total length occupied by the mixing elements ($= N_{tot}L_a$).
N_{LHE}	Total number of elements with left-hand twist.
N_{RHE}	Total number of elements with right-hand twist.
N_{tot}	Total number of elements ($= N_{RHE} + N_{LHE}$).
P	Static pressure along a cross section.
Re	Reynolds number.
t	Thickness of an element.
x	Coordinate along axial direction.
X_{add}	Mass fraction of the additive.
X_{mod}	Modified mass fraction (explained further in the text).
Z	$\Delta P_{mixer}/\Delta P_{pipe}$.
$_{add}$	property pertaining to the secondary or the additive flow.
$_{main}$	property pertaining to the primary or the main flow.
$_{mixer}$	property pertaining to a static mixer.
$_{pipe}$	property pertaining to open pipe (no mixing element inserts).

usually include one or more of the following (Paul et al., 2003):

- Blending of miscible liquids.
- Dispersion of a fluid/solid in a liquid.
- Heat transfer.

While inserts of many designs are readily available, one of the most popularly used designs is the “twisted-tape” or the “helical-type” insert (Fig. 1(a)). This includes sets of several elements having alternating left- and right-hand twists; with the leading edge of each element placed perpendicular to the trailing edge of its upstream counterpart (Fig. 1(b)). In some literature, it is also referred to by its trade name, Kenics Static Mixer or KSM. Its versatility, mechanical robustness, relatively low pressure drop, and lesser susceptibility to plugging make it popular. Mainly used for laminar and creeping flows, they find wide applicability in reacting/non-reacting systems with Newtonian/non-Newtonian fluids.

When used for blending of miscible liquids, parameters of interest in such mixers are the pressure drop along its length, and the homogeneity of the blend at the exit. In general, pressure drop is described using a “Z-factor” (Eq. (1)). It is the

ratio of the pressure drop across a mixer to that over an equal length of open pipe of same diameter (no inserts). Mixing is described using a “Coefficient of Variation (CoV)” (Eq. (2)). It is the ratio of the standard deviation of additive concentration to its average along a cross section.

$$Z = \frac{\Delta P_{mixer}}{\Delta P_{pipe}} \quad (1)$$

$$CoV = \frac{\sigma_C}{C_{av}} \quad (2)$$

In industrial setups, attaining a high degree of homogeneity with minimal pressure drop is desirable. Therefore, significant work has been done to develop/analyze correlations for pressure-drop and/or mixing performance in literature (Song and Han, 2005; Hobbs et al., 1998; Kumar et al., 2008; Byrde and Sawley, 1999). A significant portion of literature also emphasizes on modifying the individual element geometry (AR, θ) to yield the best CoV-Z trade-off (Hobbs and Muzzio, 1998a; Ling and Zhang, 1995; Jaffer and Wood, 1998; Dackson and Nauman, 2007; Arimond and Erwin, 1985; Byrde and Sawley, 1999). Byrde and Sawley (1999) did a sweep of twist angles from 0° to 360° in a setup with AR = 3 at $Re = 100$; and found the conventional 180° twist element to be optimal. They quantified mixer performance by defining an efficiency metric comprising of the radius of a vortical flow structure at the exit of the last mixer, and pressure drop. Hobbs and Muzzio (1998a) utilized techniques from dynamical systems analysis, including Poincaré sections (Ottino, 1989) to develop stretching histories of fluid tracer elements and quantify mixing in terms of CoV for a low Re ($= 0.15$) flow. They found that the amount of mixing per element is independent of AR and that the most efficient setup had $\theta = 120^\circ$. A key conclusion is that an increase in the number of elements by truncating the twist angle to $< 180^\circ$ can improve mixing for a given length of mixer pipe. This is an important finding, as space/pressure-drop is a major constraint, especially in the industry. Ling and Zhang (1995) also used Poincaré sections and tracer elements but in a helical coordinate system to conclude that 9 standard elements (AR = 1.5, $\theta = 180^\circ$) performed the best in terms of mixing as compared to shorter elements. 7 short elements (AR = 0.6, $\theta = 180^\circ$) yielded similar mixing but with a reduced pressure-drop. Dackson and Nauman (2007) (AR = 0.5, $\theta = 90^\circ$) and Arimond and Erwin (1985) ($\theta = 80^\circ$) also found similar trends with short and/or truncated elements. “Short” refers to elements with reduced AR while “truncated” refers to elements which have $\theta < 180^\circ$. Utilization of either, or a combination of them can enable inclusion of more number of elements for a given length of mixer pipe, and thus, in theory, improve mixing by increasing the number of splits of the flow.

In the aforementioned studies, the flow entering the first mixer is perfectly axial in direction and coaxial with the pipe. Therefore, the optimization of the static mixer element has been performed for such idealized scenarios. In industrial setups, the additive or the secondary fluid is introduced using beveled pipes or 90° elbows which protrude into the main pipe (also known as “Stingers”) (Fig. 2). Due to influence of the main flow as it curves around the stinger, the additive flow entering the first element may not remain perfectly coaxial with the main pipe. In other words, the incoming additive flow may be influenced by the low pressure region created as the main fluid goes downstream of the stinger.

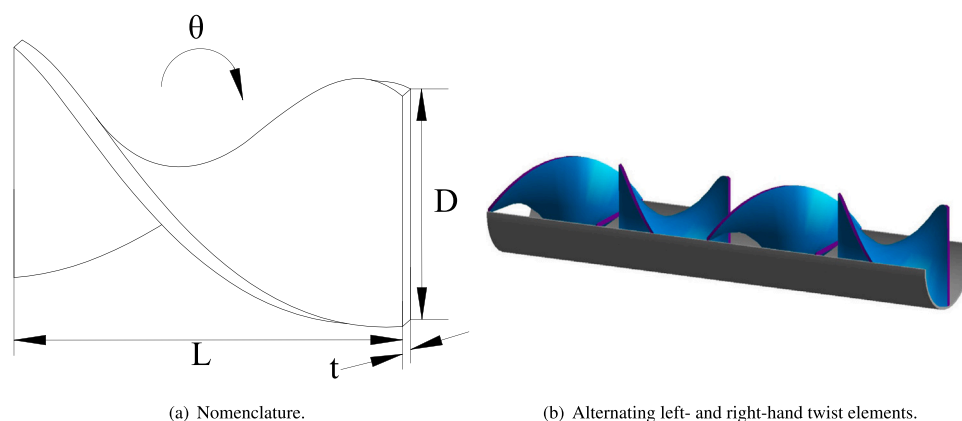


Fig. 1 – Twisted-tape mixing elements.

This flow structure is similar to the canonical case of wake formation in a cross-flow around a circular cylinder. Additionally, buoyancy effects (if $\rho_{main} \neq \rho_{add}$) may cause local offsets to the additive flow such that it does not remain coaxial with the pipe. To be representative of real-world scenarios, any optimization of the twisted-tape element must account for inexactness, offsets, and disturbances due to unintended flows introduced by injector geometry. The present work is a study to establish an optimal input parameter space (AR , θ , Re) for the operation of static mixers. By focusing on the system-level instead of the individual element-level, it accounts for the effect of unintentional flows due to the injector geometry. It also takes into consideration the influence of the injection of the additive.

Most of the work on KSM's is numerical in nature (Song and Han, 2005; van Wageningen et al., 2004; Ling and Zhang, 1995; Arimond and Erwin, 1985; Byrde and Sawley, 1999; Dackson and Nauman, 2007; Hobbs et al., 1998; Kumar et al., 2008; Hobbs and Muzzio, 1998a) due to the inherent complexity of geometry which makes non-/minimally-intrusive experimental analyses (Berkman and Calabrese, 1988; van Wageningen et al., 2004; Jaffer and Wood, 1998) difficult. Even for numerical analyses, generation of a good quality mesh is an arduous task due to the contorted geometry. Therefore, researchers have had to resort to simplifications or circumventions such as:

- zero-thickness mixer walls (Song and Han, 2005; Byrde and Sawley, 1999)
- using a fairly coarse grid (Hobbs and Muzzio, 1998b)
- periodic boundary conditions (ignoring flow fields at element junctions) (van Wageningen et al., 2004; Ling and Zhang, 1995)

- two-dimensional analysis (Arimond and Erwin, 1985)
- $Re = 0$ (Dackson and Nauman, 2007)

Mixing in the existing literature has often been quantified through either passive tracer elements (Hobbs and Muzzio, 1998a; Ling and Zhang, 1995; Hobbs and Muzzio, 1998b; Byrde and Sawley, 1999), or by additives injected through an inlet which is coaxial with the pipe (Jaffer and Wood, 1998). In this study, mixing has been quantified by analyzing the dissolution of the additive into the main flow with a finite mass diffusivity (and not by tracer particles).

2. Numerical model

2.1. Geometry, boundary conditions, and parameter space

The base geometry is constructed to be representative of a typical static mixer used in the industry as shown in Fig. 2. The main pipe has a diameter of $D = 50\text{mm}$ and the axis of the stinger is located $1D$ downstream of the main flow inlet such that their axes intersect perpendicularly. The first mixing element is situated further $1D$ downstream of the stinger and oriented as shown in Fig. 2. Adjacent elements are constructed and oriented as described in Section 1. Elements are modeled with a finite thickness of $t = 0.05D$. The total number of elements, N_{tot} , for individual computational runs are decided as described further in this section. After the last element, a length of $2D$ of open pipe is provided to isolate the parameters at the exit of the last element from any backflow effects. As used in this work, $AR = L/D$ where L is the pitch of the element and not its actual length (L_a). If $\theta \neq 180^\circ$, then $L \neq L_a$.

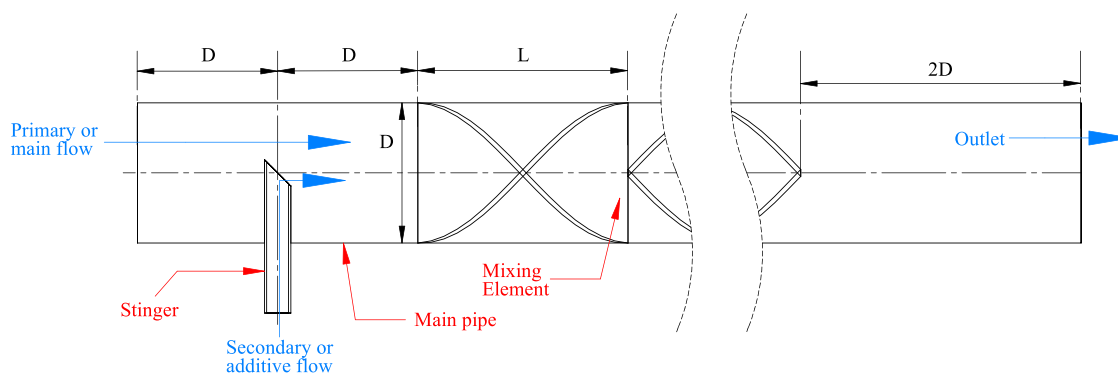


Fig. 2 – “Stinger” or additive injector used in industrial setups (for this study, $D = 50\text{mm}$).

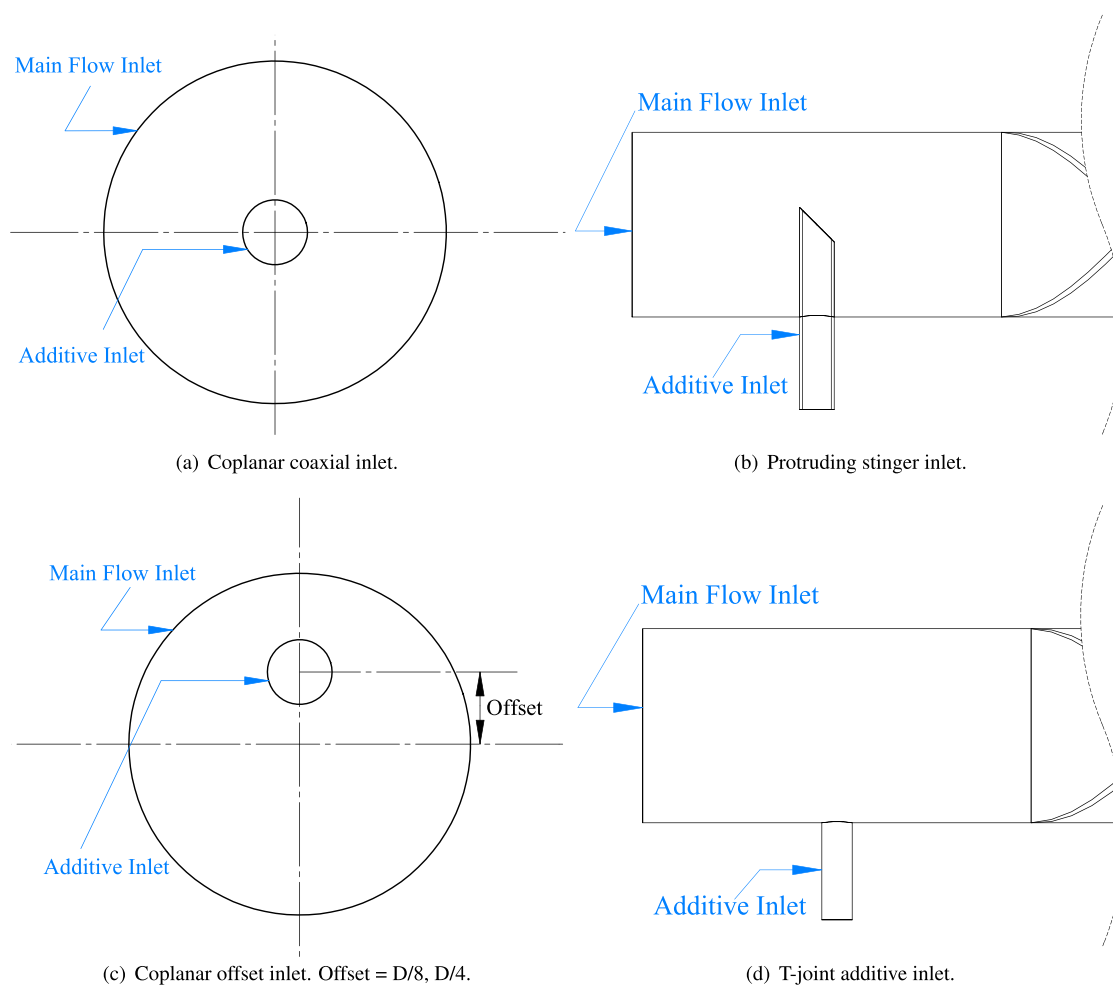


Fig. 3 – Different additive injection designs explored in this study (not to scale).

The study was conducted by varying some geometric and flow parameters. Reynolds numbers were selected based on what is commonly encountered in the petrochemical and plastic industries for mixing of highly laminar flows. Similarly, aspect ratios of 1.2 and twist angles of 180° are also ubiquitous in such industries. The other values for AR represent a shorter and a longer element as compared to $AR = 1.2$; while for twist angle, the other value represents a truncated element. The sweep of various additive inlet designs is a novel aspect of this work. The parameters are listed as follows:

- $AR = 0.6, 1.2, 1.6$
- $\theta = 140^\circ, 180^\circ$
- $Re = 1, 10, 100$
- Five different additive inlet designs as shown in Fig. 3.

The diameter of the additive inlet is kept such that the average velocity of the flow through it is the same as that from the main flow inlet. Five different additive injection designs were explored (Fig. 3):

1. Coplanar and coaxial with main flow inlet (“Coaxial”) shown in Fig. 3(a).
2. Protruding beveled pipe (“Stinger”) shown in Fig. 3(b).
3. Coplanar but axis offset with main flow inlet (“Offset”) shown in Fig. 3(c).
 - offset = $D/8$
 - offset = $D/4$

4. Non-protruding/flush pipe (“Tee-joint”) shown in Fig. 3(d).

For this study, a setup with $N_{tot} = 6$, $\theta = 180^\circ$, and $AR = 1.2$ is treated as the base case. As mentioned earlier, in addition to pressure drop, the length of a mixer is a design constraint for industrial mixers. Therefore, setups with varying θ and AR are constructed such that the pipe accommodates a whole number of mixing elements with the net length being approximately the same as the base case as listed in Table 1.

2.2. Mathematical modeling

A commercially available finite volume code, Ansys Fluent 21.1, was used for simulating the steady-state operation of the mixer under conditions listed in Section 2.1. There is some debate on the Reynolds number¹ threshold for turbulent transition in twisted-tape mixers. Values as low as $Re = 150$ (Fang and Lee, 2001) and $Re = 300$ (van Wageningen et al., 2004) have been suggested; which are much lower than that for open pipe flows of $Re \approx 2300$. Nonetheless, the values here, i.e., $Re \leq 100$ are below the aforementioned thresholds. Therefore, the flow was treated as laminar in this study. The main flow and additive fluids have the same physical properties of $\rho = 1000 \text{ kg/m}^3$ and $\mu = 50000 \text{ cP}$, and are miscible with a mass diffusivity, $D_i = 10^{-10} \text{ m}^2/\text{s}$. This value was selected as it is near the lower limit of typical values for low molecular-

¹ All Reynolds number values based on open-pipe diameter.

Table 1 – Number of mixing elements and difference in L_{tot} with respect to the base case.

AR	$\theta = 180^\circ$		$\theta = 140^\circ$	
	N_{tot}	ΔL_{tot} (%)	N_{tot}	ΔL_{tot} (%)
0.6	12	0.00	15	–2.8
1.2	6	0.00	8	+3.7
1.6	4	–11.1	6	+3.7

weight liquid pairs of 10^{-9} – $10^{-10} m^2/s$. The inlet mass flow rate from the stinger is 2.5 % of that from the main flow inlet. This value is selected as it is typical for industrial setups where an additive fluid is mixed with a bulk fluid. It was also ascertained that the results held true for fixed Reynolds numbers even with significant variations of input parameters consisting of pipe diameter and main fluid viscosity as shown in Fig. 4. Second-order discretization is used for all quantities except pressure for which PRESTO! is used as it is suitable for flows in strongly curved domains. SIMPLE algorithm is utilized to treat pressure-velocity coupling.

It must be noted that laminar mixing of scalars suffers from numerical diffusion, and successful mitigation remains an open problem in literature. However, the endeavor here has been to have a relative study by establishing a baseline for like-to-like comparison. While experiments may demonstrate a lesser degree of mixing, or higher CoV, than

what is reported here, the qualitative trends are expected to hold.

2.3. Mesh and validation

The mesh generated is of a “poly-hexcore” design as illustrated in Fig. 5. It has polyhedral elements close to boundary surfaces while the interior core of the fluid domain has hexahedral elements. A mesh-independence study was performed for the geometry with $\theta = 180^\circ$ and AR = 1.6. This particular geometry was selected as lower AR geometries require finer elements close to the mixer wall due to more angular twist per axial length, i.e., higher $d\theta/dx$. That is, all other geometries have higher cell counts. The parameters of interest were those used by Song and Han (2005) (Eq. (3)) and the results are shown in Fig. 6(a). It can be seen that even with significant increase in mesh density, the results remained nearly identical. The average mesh density of 0.65 nodes/mm³ corresponded to ~15 node points across the pipe diameter in the open pipe section. Appropriate mesh refinements in near-wall regions and in areas of expected high gradients result in average mesh density between 0.65 and 1.5 nodes/mm³ for other geometries depending on the AR. It was ensured that the minimum orthogonal quality was > 0.20 and the maximum aspect ratio was < 50 for all the meshes.

To ascertain the validity of the numerical model, comparisons are made against data available in literature. Due to

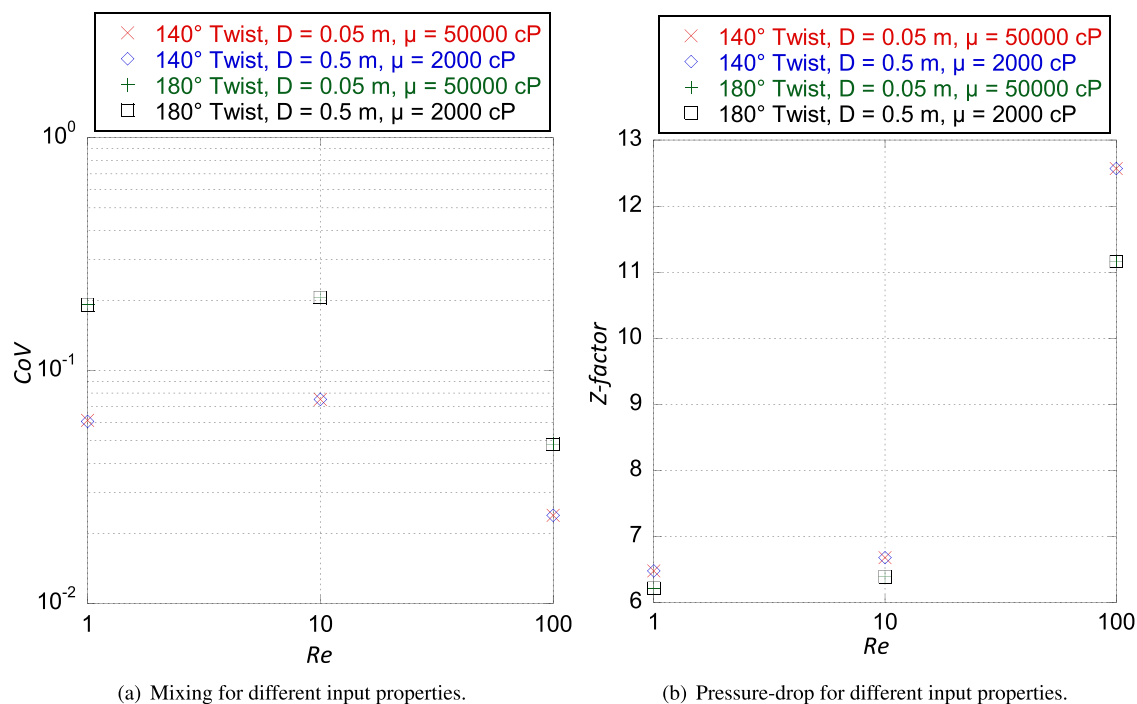


Fig. 4 – Results for fixed Reynolds numbers with significant variations in input properties.

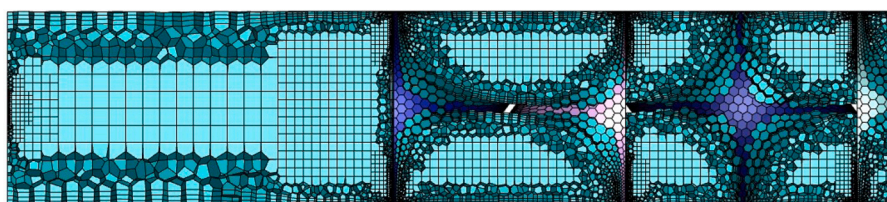


Fig. 5 – A cross sectional snapshot of the mesh for AR = 1.2, $\theta = 180^\circ$ design showing the first two elements.

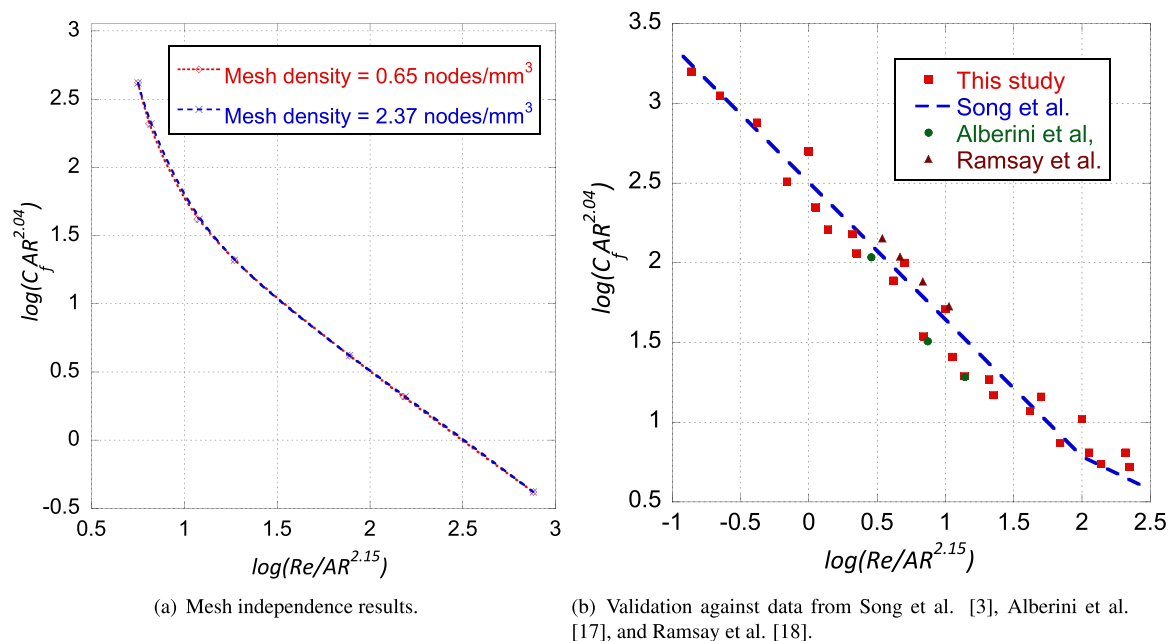


Fig. 6 – Mesh independence and validation. (a) Mesh independence results. (b) Validation against data from Song and Han (2005), Alberini et al. (2014), and Ramsay et al. (2016).

the problem of numerical diffusion discussed above, validation was performed against only pressure drop data as has been the trend in literature. The choice of results from literature against which the validation was done was based on the fact that the authors, Song and Han (2005), had performed sweeps of Re and AR ; with the numerical values encompassing the ranges used here. By doing so, they had proposed a correlation to predict the pressure drop in terms of C_f described in Eq. (3). Most variability is expected to be introduced with changing aspect ratio as $d\theta/dx$ changes; and the influence of AR is a prominent result presented in this work. Validation has also been performed against some experimental work from literature (Alberini et al., 2014; Ramsay et al., 2016).

$$C_f AR^{2.04} = K (Re/AR^{2.15})^n \quad (3)$$

Here, K and n are curve fitting parameters which are different for different ranges of the quantity in parentheses in Eq. (3). The current numerical model shows good agreement with the correlation (Fig. 6(b)) as well as experimental data of Alberini et al. (2014) and Ramsay et al. (2016). While this agreement was seen for ranges beyond what is plotted in Fig. 6(b), the axes are truncated to values which are relevant for the parameter space studied here (see Section 2.1). The slight deviations can be explained by the fact that Song and Han (2005) used zero thickness mixer walls and that K and n were optimized based on a dataset that itself had some scatter.

3. Results and discussion

The CoV – Z plots presented in this section have power regression trendlines. In addition to CoV and Z, another parameter, namely, X_{mod} is used for post-processing analysis in this work. It is based on the mass fraction of the additive as defined in Eq. (4) and has been described to better visualize low additive mass fraction regions with concentrations < 3.5%. At first glance, it may seem that the data is being clipped at an

arbitrary point. However, this value is selected as it is slightly higher than the global mass weighted average flow rate of the additive –2.5 % of the main flow. As the mass flow rate of the additive is much smaller than the main flow, color/contour plots were dominated by one of the extrema. This, in turn, did not provide sufficient visual resolution on the lower end of the scale which was of interest.

$$X_{mod} = \text{Min}(X_{add}, 3.5\%) \quad (4)$$

For a given inlet design, the CoV-Z plots for $Re = 1$ and 10 are fairly similar while those for $Re = 100$ are somewhat different as shown in Figs. 7, 8, 10, and 11. This common behavior observed across all inlet designs is consistent with Reynolds number independent velocity fields for $Re \leq 10$ reported by Hobbs et al. (1998). As mentioned earlier, mesh counts are higher for geometries with lower AR to adequately resolve the tighter twist. Therefore, some doubts may arise on the veracity of the mixing results because of the influence of varying mesh sizes between geometries, and hence, of numerical diffusion. This is addressed by the following two arguments:

1. The geometries with the same AR for the two twist angles have very similar mesh counts, and hence similar levels of error due to numerical diffusion.
2. In Figs. 7, 8, 10, and 11, one can see the general trend of improved mixing or lower CoV with decreasing AR . If the influence of numerical diffusion was dominant, this trend would not have been seen because lower $AR \rightarrow$ higher mesh count $\rightarrow$ less numerical diffusion $\rightarrow$ poorer mixing $\rightarrow$ higher CoV. This contradiction suggests that the influence of geometry on mixing is brought out despite the dependence on mesh size.

3.1. Coaxial

For the design having the additive inlet coplanar and coaxial with the main flow inlet, increasing the number of elements

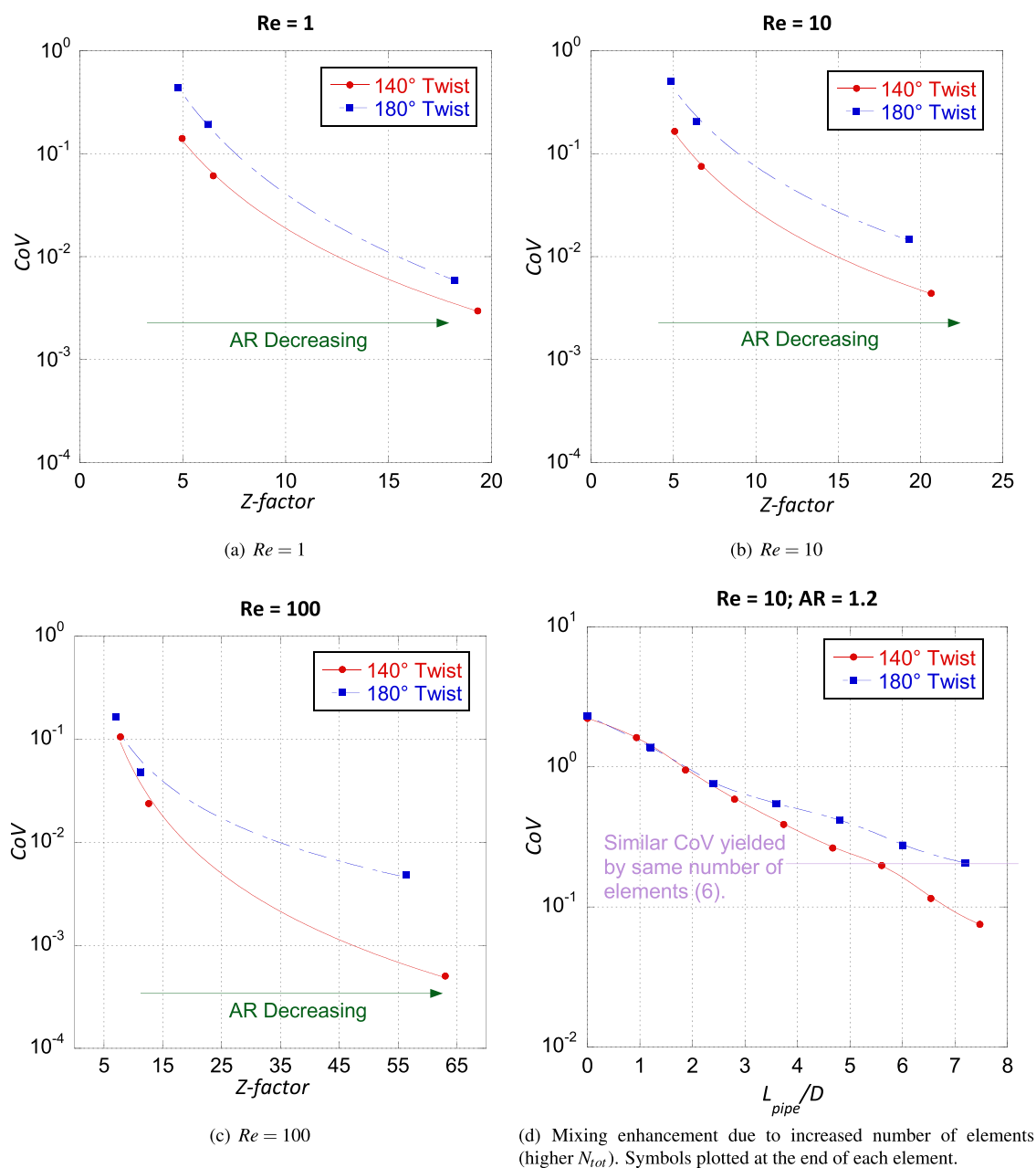


Fig. 7 – Comparison of 180° and 140° elements for Coaxial cases.

by reducing AR is seen to enhance mixing with a penalty of higher pressure drop as shown in Fig. 7. It can be thought of as more energy expended to mix the fluid streams. Fig. 7 also shows that truncated elements with twist angle of 140° perform better than their conventional 180° counterparts for a given pressure drop for all Re tested here. However, this difference is not as significant at high Re with high AR. Fig. 7(d) shows the resultant CoV is similar for the same number of elements with different θ ; and that the enhancement in mixing is caused by including more number of elements for the same pipe length. In other words, mixing improves for a given pressure drop because of an increase in the number of splits by the increased number of elements. An interim inference is: In situations where the luxury of a large increase in pressure drop is not available, mixing can be improved by utilizing truncated elements. When it is available, a combination of short and truncated elements may be used.

3.2. Stinger

In addition to the values listed in Section 2.1, a value of $AR = 0.9$ was also tested for the Stinger cases to better depict the performance trends. Unlike the Coaxial case, for $Re \leq 10$, there is an inflection point in AR beyond which truncated elements perform poorer as compared to the conventional elements. That is, they have a higher CoV for a given pressure drop as depicted in Fig. 8. This nuanced behavior makes Re a variable to be considered, in addition to pressure drop, if one is trying to increase N_{tot} for a given length of pipe. Increasing N_{tot} by lowering AR beyond a point enhances mixing for higher Re flows but it has a deleterious effect for lower Re flows.

The reason behind this behavior is revealed by studying the distribution of the additive immediately upstream of the first mixer element as shown in Fig. 9(c)–9(f) for Coaxial and Stinger injections with $AR = 0.6$, $Re = 10$, $\theta = 140^\circ$. For the

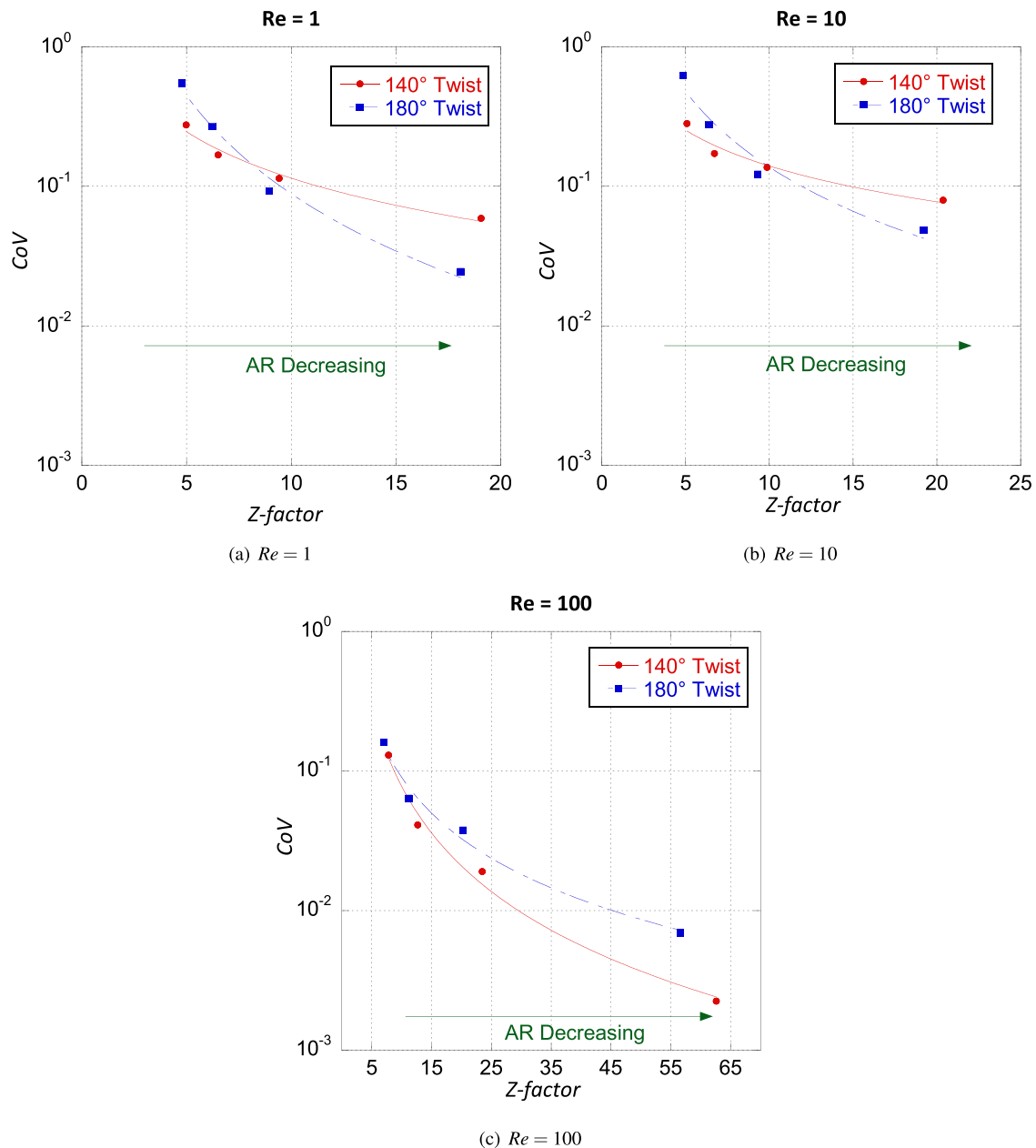


Fig. 8 – Comparison of 180° and 140° elements for Stinger cases at different Re .

Coaxial case, when the additive flow is perfectly centered within the pipe (Fig. 9(c)), the additive is evenly distributed to the left and the right sides of the element (Fig. 9(e)). Calculations of the mass flow rates of the additive in the two lobes show an almost perfect 50:50 split on mass basis, whereas, an unequal split of additive is seen for the Stinger case (Fig. 9(f)). Because of the flow around the protruding pipe, the additive is deflected away from the central axis (Fig. 9(d)). When this deflected additive flow reaches the first element it encounters more resistance on the left side than on the right side due to the right hand twist of the element; which results in a biased flow. Calculations reveal this to be an 81:19 split in the additive mass flow rate while the net mass flow rates in the lobes are equal. Thus, interplay between the flow around the pipe and the element twist creates a maldistribution of the additive which is further aggravated by the finite thickness of the elements. This goes contrary to the expectation of the flow being split into two equal halves like that in Fig. 9(c) (see Section 1 and 3.1). Furthermore, the non-ideal split at the first element cascades on and results in inhomogeneous

pockets close to the mixer and pipe walls (Fig. 9(h)) which is not seen for the ideal split (Fig. 9(g)). As for higher AR, the split is 66:34 for Stinger with $AR = 1.6$, $Re = 10$, $\theta = 140^\circ$. This is not as poor, and thus, the influence of more number of splits dominates and dictates its performance. In other words, a small inequality in split at the first element does not degrade mixer performance to such an extent that it cannot be compensated sufficiently by the presence of additional elements. Adding caveats to the interim inference from Section 3.1, it can be said that: Mixing becomes an optimization problem between increasing the number of splits with more elements and handling the unequal split at the first element.

The preceding discussion on maldistribution of additive is valid for $Re = 1$ as well. However, no such inflection point is seen for $Re = 100$ as shown in Fig. 8. The higher momentum of the main flow attenuates the influence of the flow around the protruding pipe, resulting in a less poor maldistribution of the additive - 62:38 for Stinger case with $AR = 0.6$, $Re = 100$, $\theta = 140^\circ$. Additionally, increased diffusive mixing due to higher Re ameliorates the influence of the unequal split.

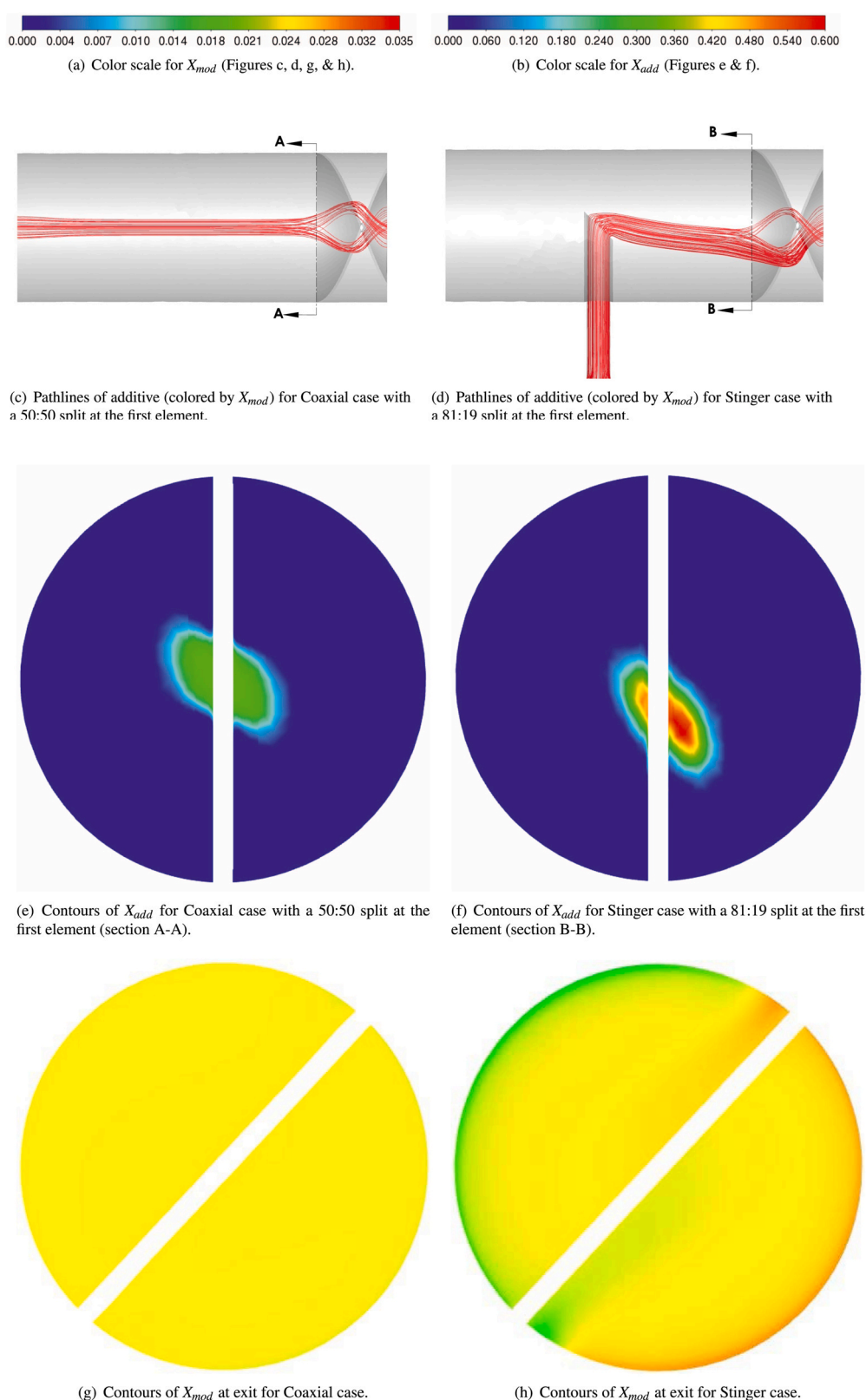


Fig. 9 – Different distributions of additive for Coaxial and Stinger additive injections while other parameters are the same ($AR = 0.6$, $Re = 10$, $\theta = 140^\circ$).

3.3. Offset

To further buttress the arguments presented in Section 3.2, two designs were tested where the additive inlet was offset by distances of $D/8$ and $D/4$ from the central axis as shown in Fig. 3(c). The aim was to check and verify if deliberately introducing an offset in the flow yielded similar results as the

Stinger case. For a small offset of $D/8$, the plots Fig. 10(a)–10(c) reveal qualitatively similar behavior as that seen for the Stinger cases. The underlying reason is also the same as before, namely, the maldistribution of additive. As for the higher offset of $D/4$, the inflection point between the two twist angles shifts to a higher AR for $Re \leq 10$ (Fig. 10(a)–10(b)). Whereas, for $Re = 100$, the two curves become coincident as

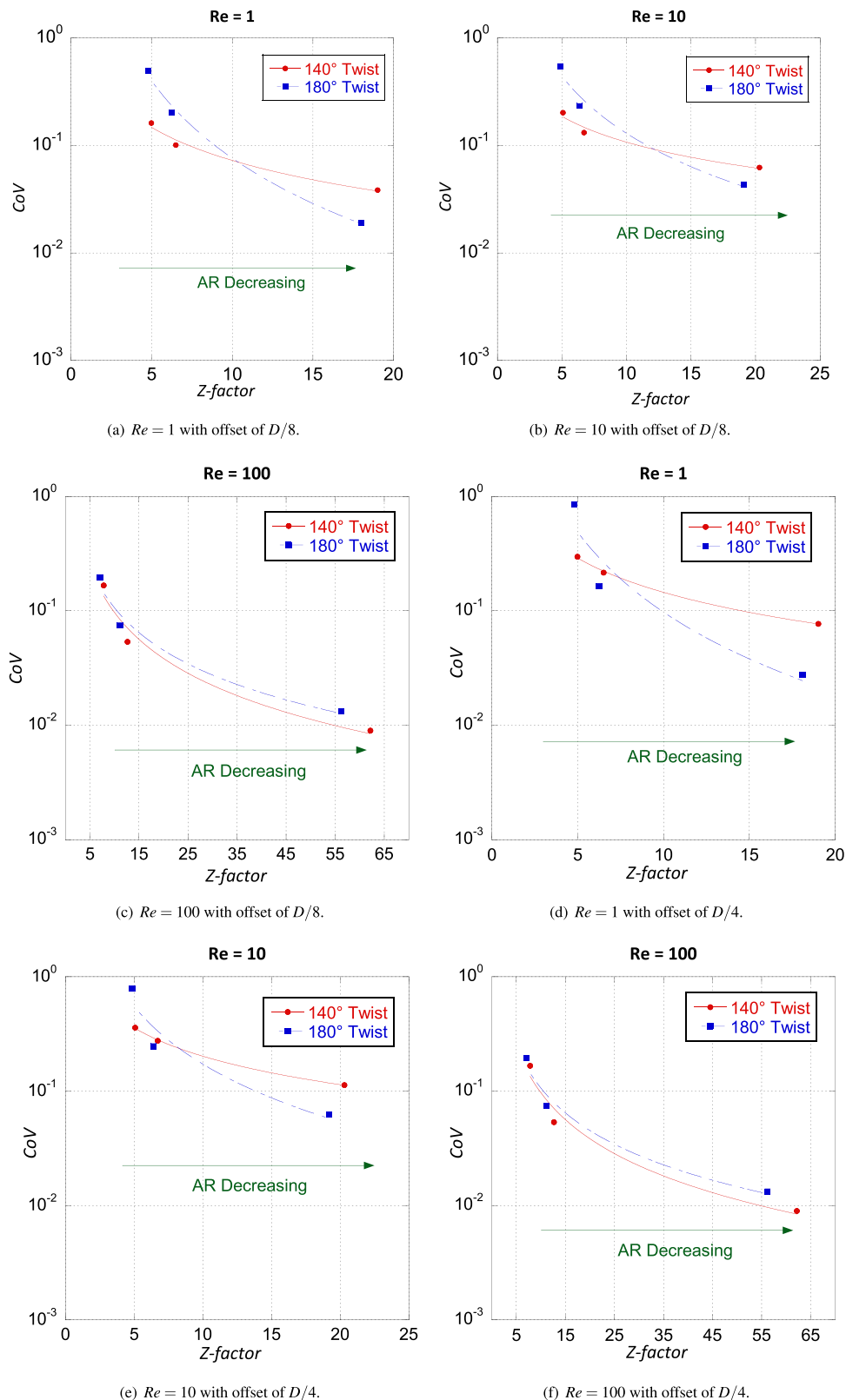


Fig. 10 – Comparison of 180° and 140° elements for Offset cases at different Re .

shown in Fig. 10(c). Thus, the complex interplay of flow offset, unequal split, and high Re renders the twist angle irrelevant for this set of parameters. It can be concluded from this section that the extent of offset away from the central axis has a major bearing on how well the element designs (θ , AR) perform in mixing. For small offsets, the behavior is similar to that for Stinger cases discussed in Section 3.2.

While the simulations were done for the main and additive flows having the same density, the results from the

Offset tests are also relevant to situations where buoyancy becomes important. Depending on the densities, in some cases buoyancy may negate the offset due to the flow around the stinger while in other cases it may exacerbate it; the magnitude being dictated by the difference of densities. It is uncommon to have setups where the densities of the additive and the main flows are exactly the same or very close to each other. Therefore, it is difficult to ensure a perfectly coaxial additive flow going into the first mixer element. Even

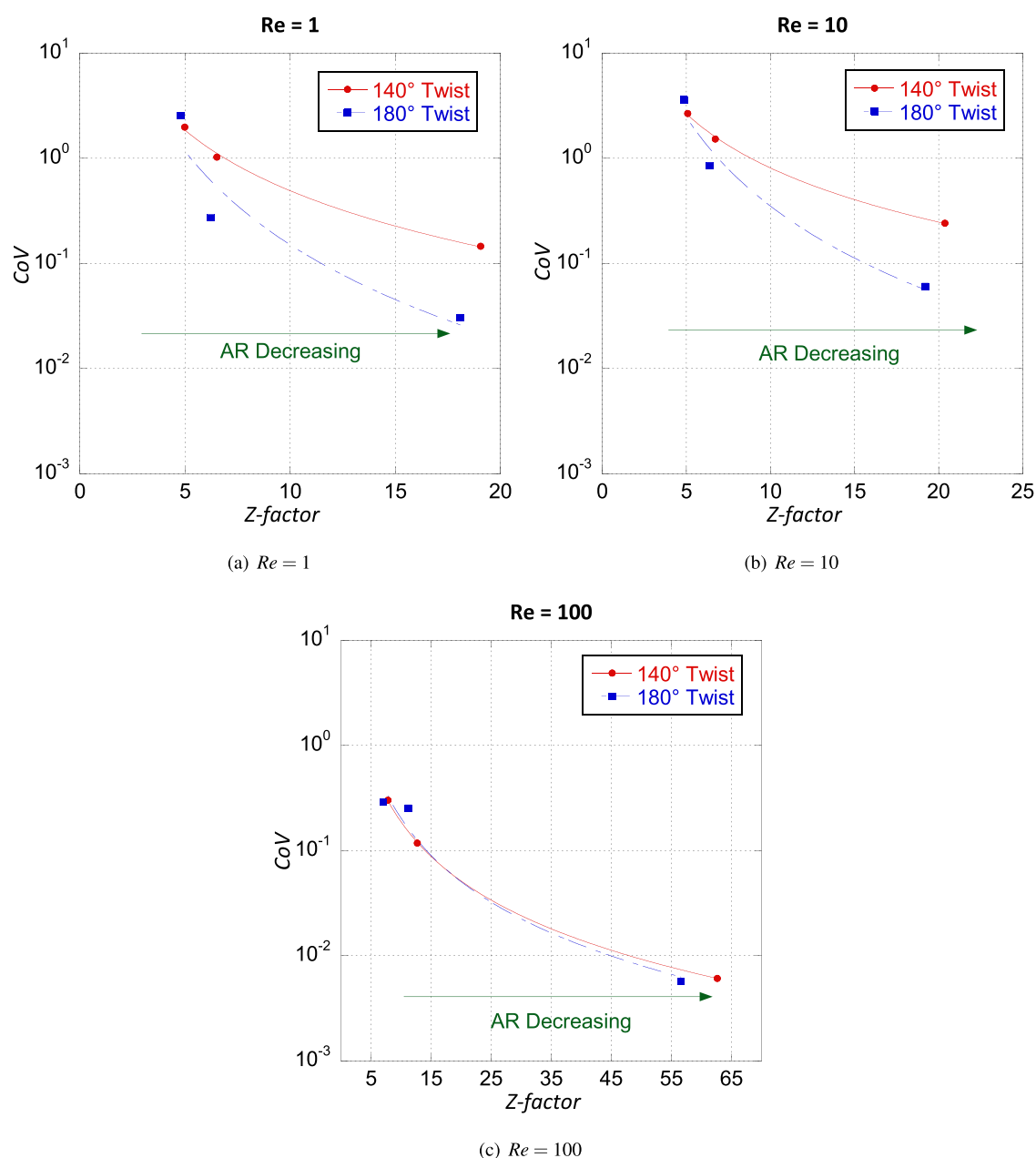


Fig. 11 – Comparison of 180° and 140° elements for Tee-joint cases at different Re .

a diligently designed mixer may show some inexactness due to the reasons mentioned above. Thus, system level design is necessitated for static mixers.

3.4. Tee-joint

As a final test, a tee-joint type inlet, as shown in Fig. 3(d), was studied. In this arrangement, the additive comes in and flows along the wall of the main pipe as can be seen in Fig. 12. It must be noted that desirable operation in tee mixers is that the cross flow penetrates into the core of the main flow and stays away from the walls (Pan and Meng, 2001). However, the endeavor here was to test a setup representative of the maximum possible offset of the additive from the central axis, i.e., $D/2$. For $Re \leq 10$, a logical continuation of the trend with increasing offset is seen via Figs. 7 & 10. That is, for the Tee-joint cases, the 180° mixer is always better than the 140° mixer except for the highest AR (Fig. 11(a)–11(b)). For $Re = 100$, the behavior (Fig. 11(c)) is same as the $D/4$ offset (Fig. 10(f)) where the two curves are coincident.

Maldistribution of the additive is equally bad for both twist angles with the split being $\approx 92:8$ for both of them. From what has been learnt in the preceding sections, with similar maldistributions, 140° elements should have performed better due to more number of splits; but that is contrary to the observation. The underlying reasons are revealed by studying the pathlines of the additive in Fig. 12. It can be observed that large volumes of the upper half of the mixer tube remain devoid of the additive for the 140° elements. On the other hand, the additive is present in a larger volumetric region of the tube for the 180° elements. The wall-bound additive is able to reach this larger region as it flows along the surface of an element with a “fuller” 180° twist. Flow along the truncated 140° twist does not penetrate into the upper portion of the mixer tube as completely. Consequently, pathlines close to the exit show more inhomogeneity for the 140° twist elements. This inhomogeneity is also reminiscent of the stratification close to the walls for the Stinger case in Fig. 9(h). Thus, it can be argued that the latter had some similar aspects of hydrodynamics. It is worth highlighting that

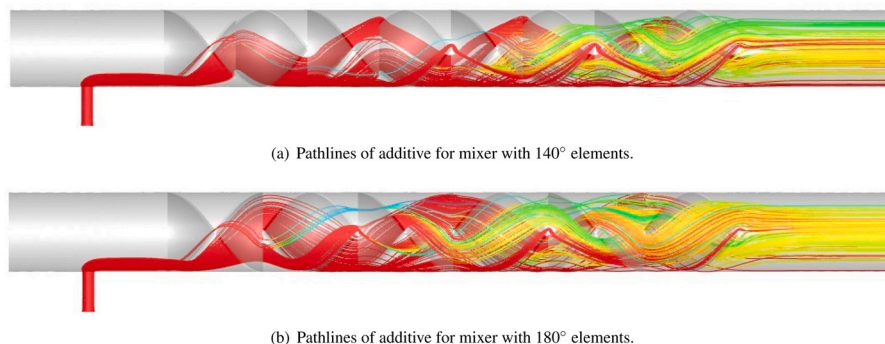


Fig. 12 – Pathlines of additive for Tee-joint with AR = 1.2 and Re = 10. For color scale, use Fig. 9(a).

with this extreme offset from the central axis, the mixing behavior is not controlled by the splitting of the flow by the mixing elements; but by the flow along its curved walls.

4. Conclusions

The necessity of system-level design for efficient twisted-tape static mixers in industrial setups is established via a parametric study. Truncated and short elements perform better than their conventional counterparts for idealized flows where the additive flow is perfectly centered in the pipe. However, the parameter space for their superior operation is curtailed for industrially relevant designs. This study determines these curtailed parametric boundaries and explains the reasons behind the aforesaid curtailment. The optimal pair of twist angle and aspect ratio is a function of Reynolds number and additive offset distance. For $Re \leq 10$, reduction of twist angle from 180° to 140° improves mixing for aspect ratios greater than approximately 0.9; whereas, for low aspect ratios, it makes mixing worse. For Reynolds number of 100, the 140° twist angle outperforms the 180° twist angle for all aspect ratios studied, i.e., higher homogeneity for similar pressure drop. All results are presented for approximately the same net length of the mixer. The performance of a static mixer is shown to be significantly determined by a complex interplay of upstream flow field, twist angle, aspect ratio, element thickness, and Reynolds number. This is especially noteworthy as some of these parameters were being approximated/not considered in previous literature. Mixing is observed to be an optimization problem between increasing the number of splits with more elements and handling the non-ideal split at the first element. In the future, the study may be expanded to broader sweeps of the parameter space and include results with non-Newtonian flows.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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